

Ultimate Complete DSA Patterns Guide

450+ Real Placement Questions (LeetCode + GFG + HackerRank)

Tier	Patterns
Tier 1 (Highest ROI)	Arrays, Strings, HashMap, Sliding Window, Two Pointers, Binary Search
Tier 2	Linked List, Stack, Queue, Heap, Greedy, Recursion
Tier 3	Graphs, Backtracking, Trie, Dynamic Programming
Mixed Patterns	Prefix Sum + HashMap, Sliding Window + HashMap, Binary Search + Greedy

Preparation Strategy

- Solve pattern-wise, not randomly.
- Focus on recognizing triggers.
- Practice Easy → Medium progression.
- Repeat failed questions.
- Time yourself during practice.
- Explain the logic aloud after solving.

Two Pointers

- 1. Two Sum II
- 2. 3Sum
- 3. 4Sum
- 4. Container With Most Water
- 5. Sort Colors
- 6. Remove Duplicates from Sorted Array
- 7. Move Zeroes
- 8. Merge Sorted Array
- 9. Valid Palindrome
- 10. Squares of a Sorted Array
- 11. Trapping Rain Water
- 12. Boats to Save People
- 13. Partition Labels
- 14. Backspace String Compare
- 15. Reverse Vowels of a String
- 16. Dutch National Flag
- 17. Pair Sum Closest
- 18. Triplet Sum Closest
- 19. Remove Element
- 20. Intersection of Two Arrays II
- 21. Shortest Unsorted Continuous Subarray
- 22. Minimum Difference Pair
- 23. Count Pairs Less Than Target
- 24. Sort Array By Parity
- 25. Reverse String
- 26. Palindrome Linked List
- 27. Linked List Cycle
- 28. Max Consecutive Ones
- 29. Merge Intervals
- 30. Interval List Intersections

Sliding Window

- 1. Longest Substring Without Repeating Characters
- 2. Minimum Window Substring
- 3. Find All Anagrams in a String
- 4. Permutation in String
- 5. Longest Repeating Character Replacement
- 6. Maximum Sum Subarray of Size K
- 7. Subarray Product Less Than K
- 8. Fruit Into Baskets
- 9. Sliding Window Maximum
- 10. Minimum Size Subarray Sum
- 11. Binary Subarrays With Sum
- 12. Max Consecutive Ones III
- 13. Count Number of Nice Subarrays
- 14. Longest Subarray of 1s After Deleting One Element
- 15. Grumpy Bookstore Owner
- 16. Repeated DNA Sequences
- 17. Diet Plan Performance
- 18. Maximum Average Subarray
- 19. Contains Duplicate II
- 20. K Radius Subarray Averages
- 21. Longest Harmonious Subsequence
- 22. Substring with Concatenation of All Words
- 23. Smallest Range Covering Elements
- 24. Minimum Swaps Together
- 25. Longest Continuous Increasing Subsequence
- 26. Maximum Points You Can Obtain from Cards
- 27. Defuse the Bomb
- 28. Equal Substrings Within Budget
- 29. Get Equal Substrings Within Budget
- 30. Longest Nice Subarray

Prefix Sum + HashMap

- 1. Subarray Sum Equals K
- 2. Continuous Subarray Sum
- 3. Pivot Index
- 4. Running Sum of 1D Array
- 5. Product of Array Except Self
- 6. Maximum Size Subarray Sum Equals K
- 7. Count Number of Nice Subarrays
- 8. Contiguous Array
- 9. Range Sum Query
- 10. Corporate Flight Bookings
- 11. Find Pivot Index
- 12. Path Sum III
- 13. Subarrays Divisible by K
- 14. Count Equal 0s and 1s
- 15. Longest Consecutive Sequence
- 16. Count Bad Pairs
- 17. Difference Array
- 18. Check if Array Pairs Are Divisible by K
- 19. Count Triplets That Can Form Equal XOR
- 20. Find the Middle Index
- 21. Prefix XOR Queries
- 22. Minimum Operations to Reduce X to Zero
- 23. Sum of Absolute Differences
- 24. Ways to Split Array
- 25. Car Pooling
- 26. Split Array Largest Sum
- 27. Find All Good Indices
- 28. Shifting Letters II
- 29. Range Addition
- 30. K-diff Pairs in an Array

Binary Search

- 1. Binary Search
- 2. Search Insert Position
- 3. Find First and Last Position
- 4. Search in Rotated Sorted Array
- 5. Find Minimum in Rotated Sorted Array
- 6. Find Peak Element
- 7. Sqrt(x)
- 8. Koko Eating Bananas
- 9. Capacity To Ship Packages Within D Days
- 10. Aggressive Cows
- 11. Allocate Minimum Number of Pages
- 12. Split Array Largest Sum
- 13. Minimum Days to Make Bouquets
- 14. Magnetic Force Between Two Balls
- 15. Single Element in a Sorted Array
- 16. Median of Two Sorted Arrays
- 17. Search a 2D Matrix
- 18. Peak Index in Mountain Array
- 19. Guess Number Higher or Lower
- 20. Arranging Coins
- 21. Perfect Square
- 22. Find K Closest Elements
- 23. Search in Nearly Sorted Array
- 24. Nth Root of Integer
- 25. Painter's Partition Problem
- 26. Ship Packages Within D Days
- 27. Successful Pairs of Spells and Potions
- 28. Maximum Candies Allocated
- 29. Minimum Speed to Arrive on Time
- 30. Find Smallest Divisor Given Threshold

Linked List

- 1. Reverse Linked List
- 2. Reverse Linked List II
- 3. Reverse Nodes in K Group
- 4. Linked List Cycle
- 5. Linked List Cycle II
- 6. Middle of Linked List
- 7. Palindrome Linked List
- 8. Reorder List
- 9. Merge Two Sorted Lists
- 10. Merge K Sorted Lists
- 11. Add Two Numbers
- 12. Remove Nth Node From End
- 13. Intersection of Two Linked Lists
- 14. Flatten Multilevel Doubly Linked List
- 15. Copy List with Random Pointer
- 16. Swap Nodes in Pairs
- 17. Rotate List
- 18. Sort List
- 19. Partition List
- 20. Odd Even Linked List
- 21. Delete Node in Linked List
- 22. Remove Duplicates from Sorted List
- 23. Convert Binary Number in Linked List
- 24. Twin Sum of Linked List
- 25. Split Linked List in Parts
- 26. Insertion Sort List
- 27. Design Linked List
- 28. LRU Cache
- 29. Browser History
- 30. My Circular Queue

Stack & Monotonic Stack

- 1. Valid Parentheses
- 2. Min Stack
- 3. Daily Temperatures
- 4. Next Greater Element I
- 5. Next Greater Element II
- 6. Largest Rectangle in Histogram
- 7. Trapping Rain Water
- 8. Asteroid Collision
- 9. Remove K Digits
- 10. Stock Span Problem
- 11. Sum of Subarray Minimums
- 12. Decode String
- 13. Evaluate Reverse Polish Notation
- 14. Basic Calculator
- 15. Generate Parentheses
- 16. Simplify Path
- 17. Online Stock Span
- 18. 132 Pattern
- 19. Car Fleet
- 20. Score of Parentheses
- 21. Maximum Frequency Stack
- 22. Baseball Game
- 23. Crawler Log Folder
- 24. Make The String Great
- 25. Remove All Adjacent Duplicates
- 26. Validate Stack Sequences
- 27. Backspace String Compare
- 28. Reverse Polish Evaluation
- 29. Expression Add Operators
- 30. Remove Duplicate Letters

Heap / Priority Queue

- 1. Kth Largest Element in an Array
- 2. Top K Frequent Elements
- 3. Find Median from Data Stream
- 4. Merge K Sorted Lists
- 5. K Closest Points to Origin
- 6. Last Stone Weight
- 7. Task Scheduler
- 8. Meeting Rooms II
- 9. Connect Ropes
- 10. Reorganize String
- 11. Smallest Range Covering Elements
- 12. IPO
- 13. Furthest Building You Can Reach
- 14. Sort Characters By Frequency
- 15. Maximum Performance of a Team
- 16. Ugly Number II
- 17. Seat Reservation Manager
- 18. Stone Game VI
- 19. Kth Smallest Element in Sorted Matrix
- 20. Find K Pairs with Smallest Sums
- 21. Total Cost to Hire K Workers
- 22. Design Twitter
- 23. Minimum Cost to Connect Sticks
- 24. Maximum Subsequence Score
- 25. Take Gifts From Richest Pile
- 26. Sliding Window Median
- 27. Top K Frequent Words
- 28. Merge Intervals
- 29. Distant Barcodes
- 30. Meeting Rooms

Greedy

- 1. Jump Game
- 2. Jump Game II
- 3. Gas Station
- 4. Candy
- 5. Assign Cookies
- 6. Partition Labels
- 7. Non-overlapping Intervals
- 8. Minimum Number of Arrows to Burst Balloons
- 9. Boats to Save People
- 10. Activity Selection
- 11. Task Scheduler
- 12. Maximum Units on a Truck
- 13. Lemonade Change
- 14. Best Time to Buy and Sell Stock II
- 15. Bag of Tokens
- 16. Can Place Flowers
- 17. Queue Reconstruction by Height
- 18. Wiggle Subsequence
- 19. Hand of Straights
- 20. Minimum Platforms
- 21. Fractional Knapsack
- 22. Job Sequencing Problem
- 23. Rescue Boats
- 24. Split a String in Balanced Strings
- 25. Longest Palindrome
- 26. Erase Overlap Intervals
- 27. Minimum Add to Make Parentheses Valid
- 28. Reduce Array Size to Half
- 29. Maximum 69 Number
- 30. Integer Replacement

Recursion & Backtracking

- 1. Subsets
- 2. Subsets II
- 3. Permutations
- 4. Permutations II
- 5. Combination Sum
- 6. Combination Sum II
- 7. Letter Combinations of a Phone Number
- 8. Generate Parentheses
- 9. Palindrome Partitioning
- 10. Restore IP Addresses
- 11. N Queens
- 12. Sudoku Solver
- 13. Word Search
- 14. Rat in a Maze
- 15. M Coloring Problem
- 16. Beautiful Arrangement
- 17. Expression Add Operators
- 18. Letter Tile Possibilities
- 19. Matchsticks to Square
- 20. Combination Sum III
- 21. Split String Into Descending Consecutive Values
- 22. Unique Paths III
- 23. Word Break II
- 24. Gray Code
- 25. Binary Watch
- 26. Permutations of String
- 27. K-th Symbol in Grammar
- 28. Tower of Hanoi
- 29. Generate Binary Strings
- 30. Josephus Problem

Graphs

- 1. Number of Islands
- 2. Clone Graph
- 3. Course Schedule
- 4. Word Ladder
- 5. Rotting Oranges
- 6. Pacific Atlantic Water Flow
- 7. Surrounded Regions
- 8. Graph Valid Tree
- 9. Shortest Path in Binary Matrix
- 10. Is Graph Bipartite
- 11. Flood Fill
- 12. 01 Matrix
- 13. Walls and Gates
- 14. Alien Dictionary
- 15. Network Delay Time
- 16. Cheapest Flights Within K Stops
- 17. Dijkstra Algorithm
- 18. Detect Cycle in Graph
- 19. Topological Sort
- 20. Kosaraju Algorithm
- 21. Prim's Algorithm
- 22. Kruskal Algorithm
- 23. Accounts Merge
- 24. Evaluate Division
- 25. Number of Provinces
- 26. Find Eventual Safe States
- 27. Redundant Connection
- 28. Open the Lock
- 29. Minimum Genetic Mutation
- 30. Word Search II

Dynamic Programming

- 1. Climbing Stairs
- 2. House Robber
- 3. House Robber II
- 4. Coin Change
- 5. Longest Increasing Subsequence
- 6. Longest Common Subsequence
- 7. 0/1 Knapsack
- 8. Partition Equal Subset Sum
- 9. Edit Distance
- 10. Decode Ways
- 11. Unique Paths
- 12. Minimum Path Sum
- 13. Target Sum
- 14. Best Time to Buy and Sell Stock with Cooldown
- 15. Word Break
- 16. Palindromic Substrings
- 17. Longest Palindromic Substring
- 18. Maximum Product Subarray
- 19. Jump Game
- 20. Combination Sum IV
- 21. Perfect Squares
- 22. Delete and Earn
- 23. Distinct Subsequences
- 24. Interleaving String
- 25. Cherry Pickup
- 26. Triangle
- 27. Burst Balloons
- 28. Frog Jump
- 29. Paint House
- 30. Dungeon Game